

1.7 Empirical Applications

Growth Accounting

Growth accounting helps us understand the factors driving economic growth. It breaks down growth into contributions from various elements like labor, capital, and technological progress. This method was developed by economists Abramovitz and Solow in the 1950s.

To see how it works, we use a production function to look at how capital, labor, and technology impact output. By measuring how much each factor contributes, we can calculate the "Solow residual," which often represents technological progress or other unexplained growth factors.

While growth accounting shows the immediate causes of growth, like capital accumulation and improved labor, it doesn't explain why these factors change over time. For instance, rapid growth in East Asian countries was largely due to increased investment and better labor, rather than technological advancements. Some economists debate whether capital accumulation alone explains growth or if other factors play a bigger role.

Growth accounting has also been used to study productivity trends, such as the slowdown in U.S. productivity in the 1970s and its rebound in the 1990s, largely due to the impact of information technology like computers. Initially, computers didn't significantly boost productivity, but as they spread and industries adjusted, they became a major factor in productivity growth after the mid-1990s.

Convergence

The question of whether poor countries grow faster than rich ones has gained attention. There are three main reasons why we might expect this to happen:

1. According to the Solow model, countries move toward their balanced growth paths, so poorer countries should catch up to richer ones over time.
2. The Solow model also suggests that countries with more capital (machines, buildings) have lower returns on capital. This encourages capital to flow from rich to poor countries, promoting convergence.
3. Poor countries might lag in adopting new technologies, but as they catch up and start using the latest methods, income differences could shrink.

Baumol (1986) studied this by analyzing the growth of 16 industrialized countries from 1870 to 1979. His findings suggested almost perfect convergence, meaning poor countries grew faster than rich ones. However, DeLong (1988) pointed out issues with Baumol's results. First, the sample only included countries that had long data histories, which biased the findings toward faster-growing poor countries. DeLong added more countries to the sample and found weaker evidence for convergence. Second, measurement errors in estimating income levels from 1870 also caused bias, making it look like poor countries were catching up faster than they really were.

Overall, when correcting for these problems, the case for convergence weakens significantly, and there is little evidence to suggest poor countries consistently grow faster than rich ones.